Machine Learning Fundamentals

Introduction to Machine Learning

Machine Learning (ML) is a branch of artificial intelligence that enables computers to learn from data and improve their performance without being explicitly programmed. Unlike traditional programming, where instructions are written explicitly, ML algorithms learn rules directly from data.

Key Concepts

- **Training Data**: Examples the model learns from.
- **Algorithms**: Mathematical methods that learn from data.
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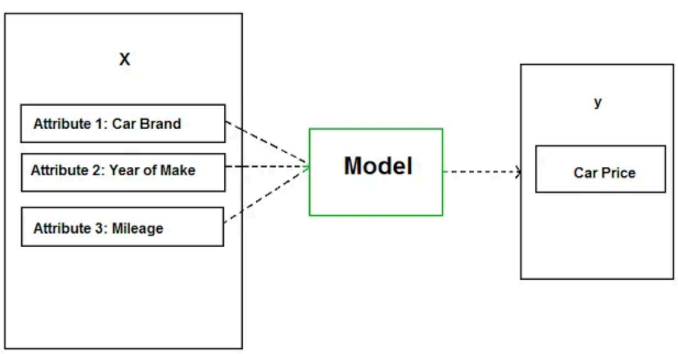
Types of Machine Learning

Machine learning methods are commonly grouped into **three main types** based on how they learn from data: **Supervised**, **Unsupervised**, and **Reinforcement Learning**. Each type addresses a different class of problems and learning scenarios.

1. Supervised Learning

Supervised learning involves algorithms that learn from labeled datasets, meaning each data point has an associated output. The goal is to learn a general rule mapping inputs to outputs, so that the model can predict the correct output for new, unseen inputs. Essentially, the model is "supervised" by the known answers during training.



Supervised Learning Model

Source: Medium

A- Classification

Classification – Predicting a **category** or class label for each input.

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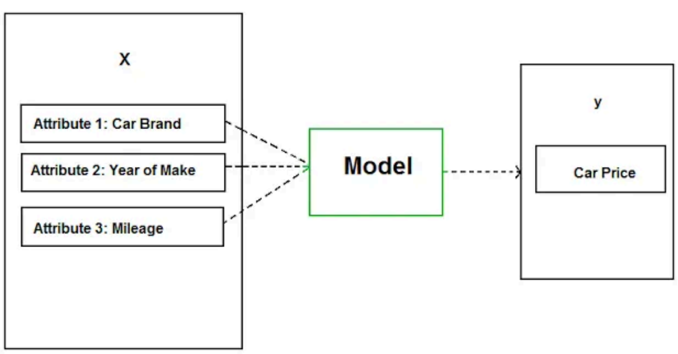

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b. <u>Classification Models</u>

- **Logistic Regression**
  - **Why:** Simple, interpretable, fast training.
  - **Best for:** Binary classification tasks with linearly separable data (e.g., spam detection, credit approval).
- **Random Forest Classifier**
  - **Why:** Handles large datasets, reduces overfitting, good accuracy.
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  - **Why:** Effective with clear decision boundaries, robust against overfitting.
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B- Regression

Regression – Predicting a **continuous numeric value** for each input.

a. <u>Examples:</u>

- **House Price Prediction:** Estimating property prices based on features like location and size.
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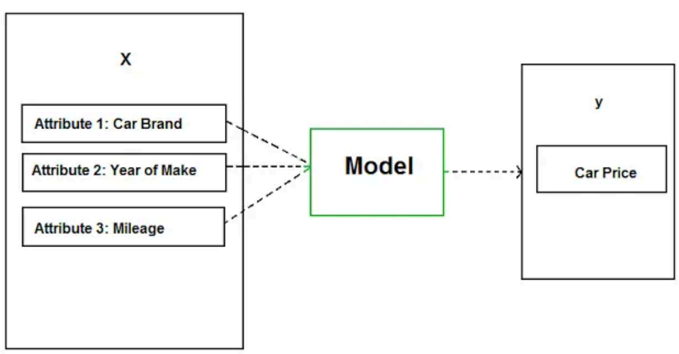

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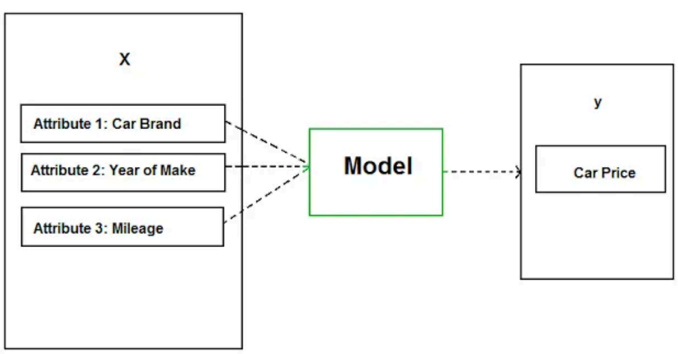

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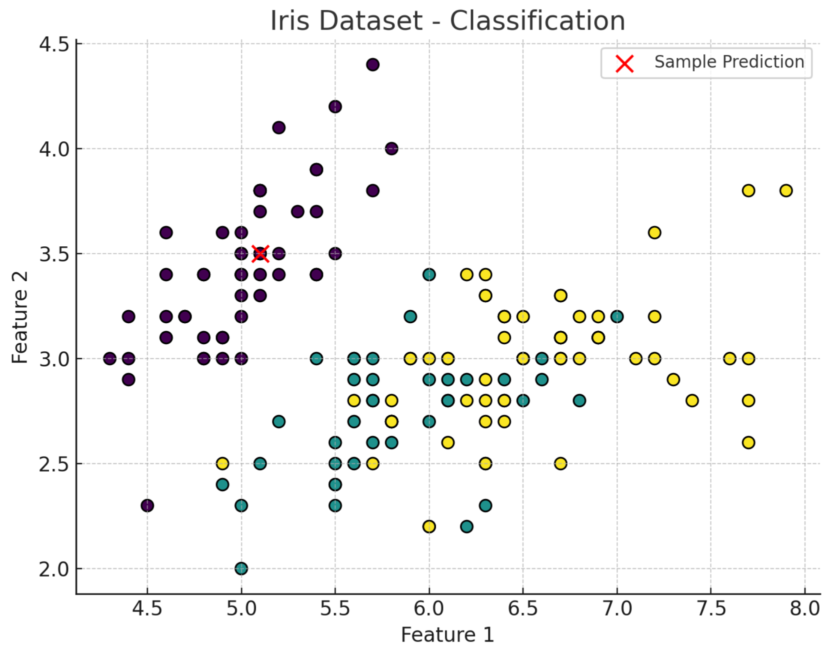


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sample = X[0]
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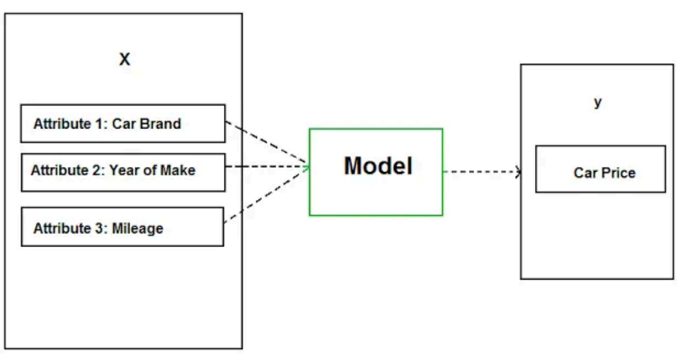

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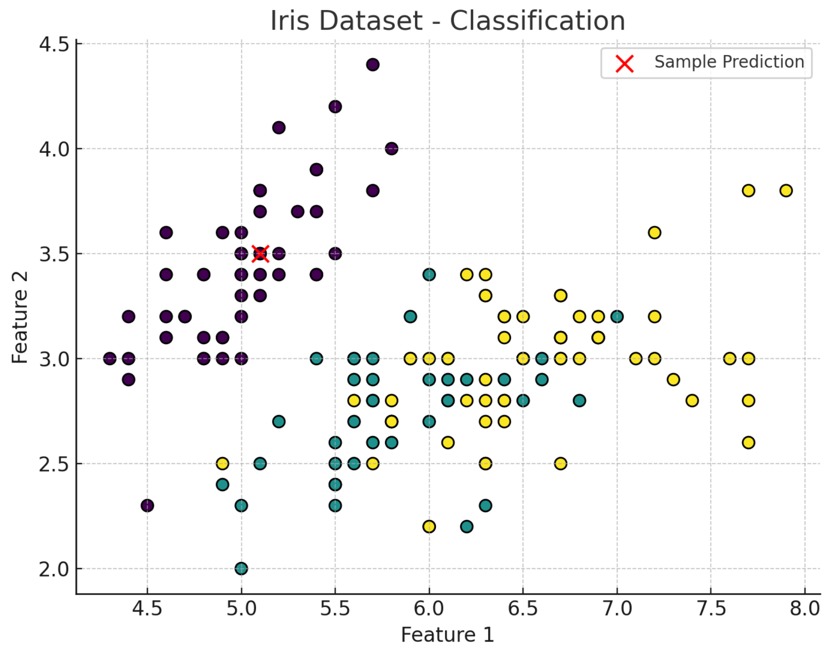


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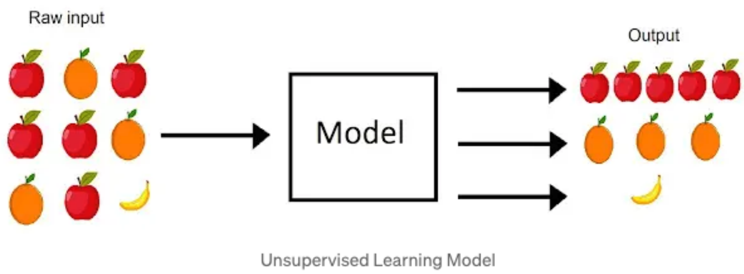

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  - **What:** Neural networks trained to compress and reconstruct data, capturing meaningful patterns.

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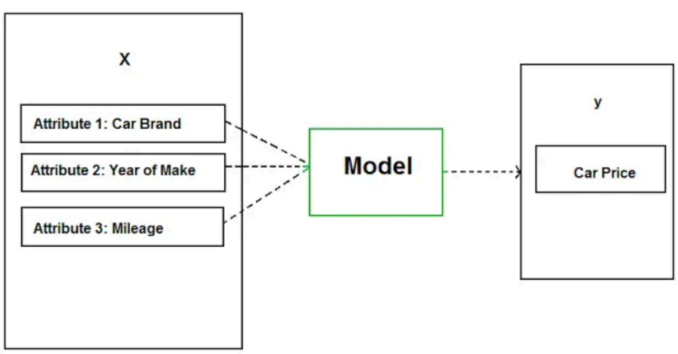

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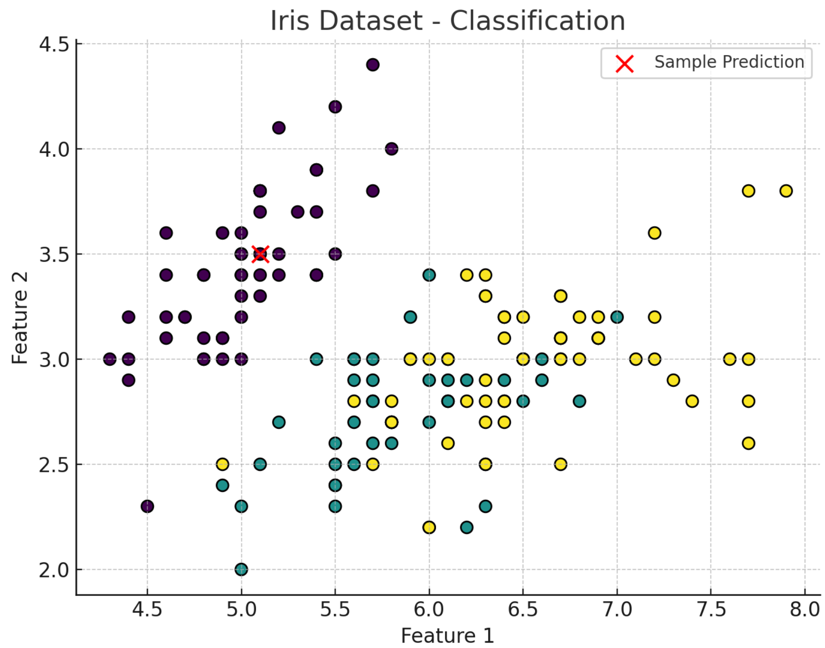


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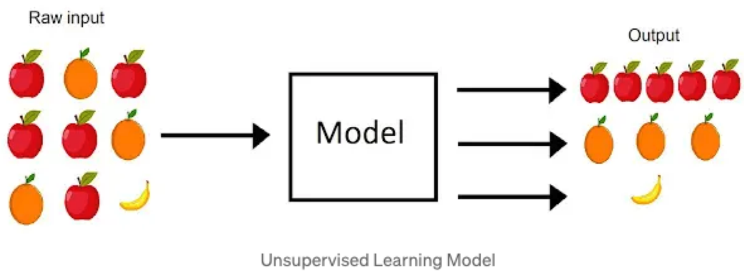

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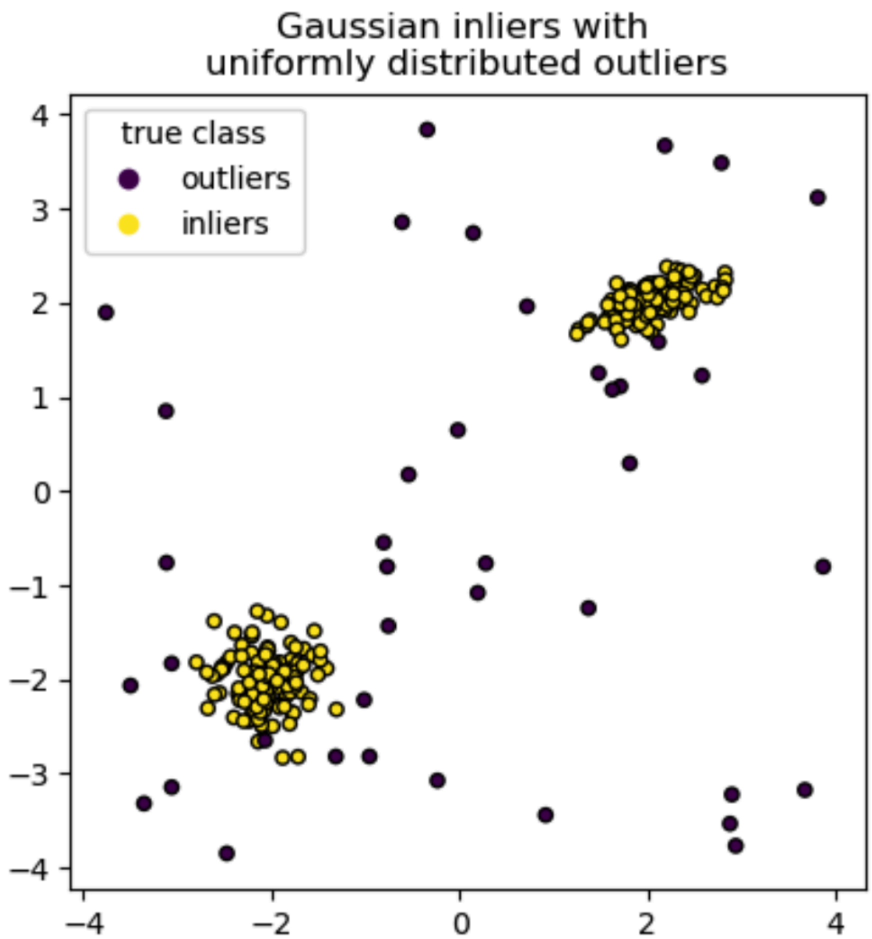

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Machine learning methods are commonly grouped into **three main types** based on how they learn from data: **Supervised**, **Unsupervised**, and **Reinforcement Learning**. Each type addresses a different class of problems and learning scenarios.

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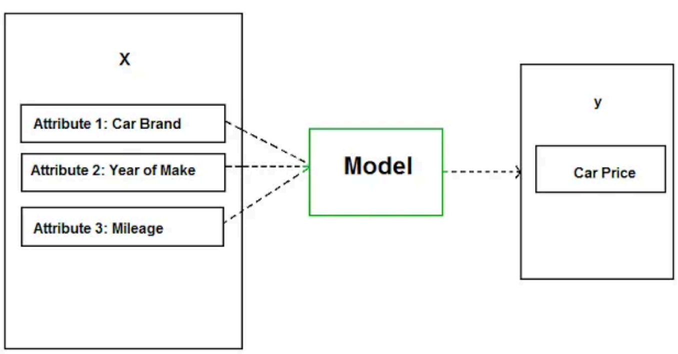

Supervised Learning Model

Source: Medium

A- Classification

Classification – Predicting a **category** or class label for each input.

a. Examples:

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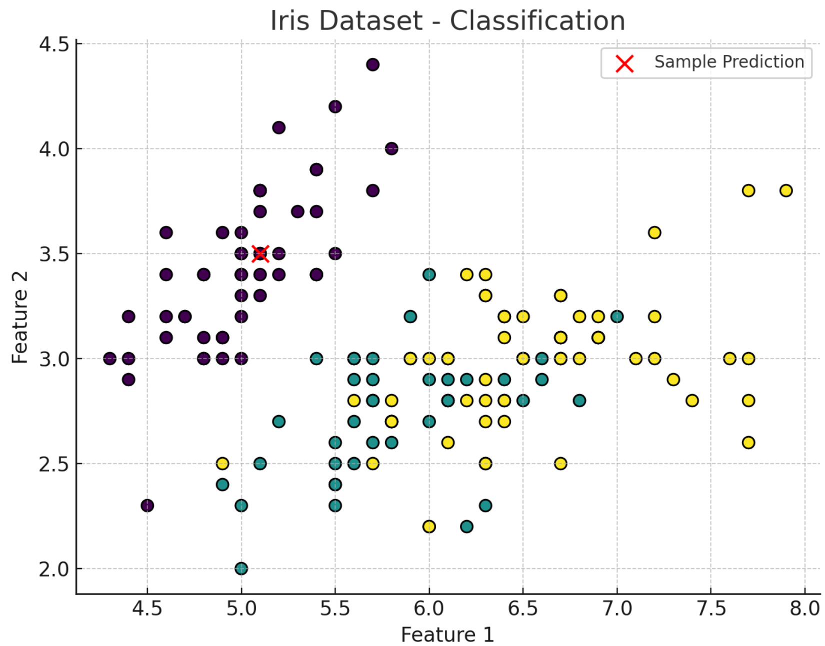


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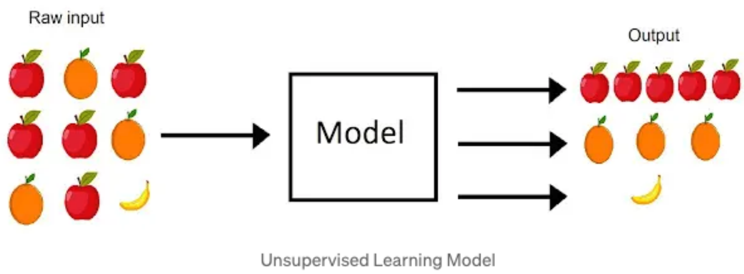

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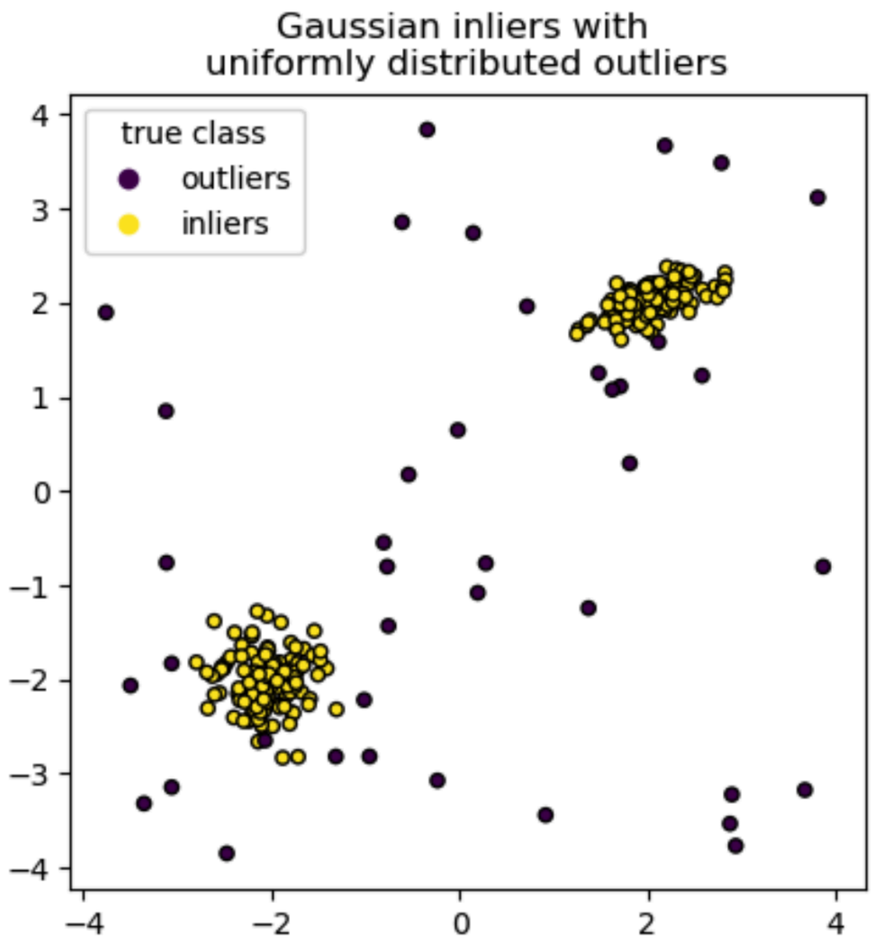

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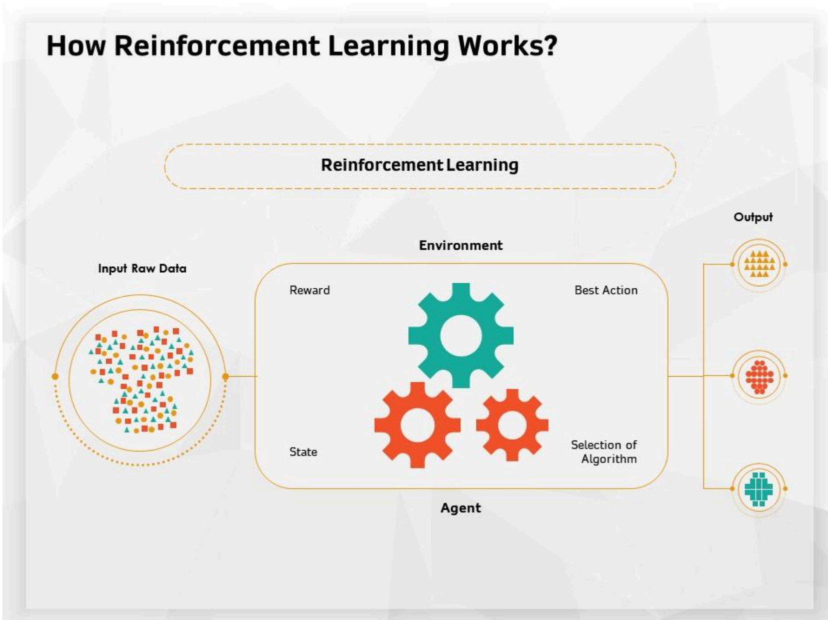

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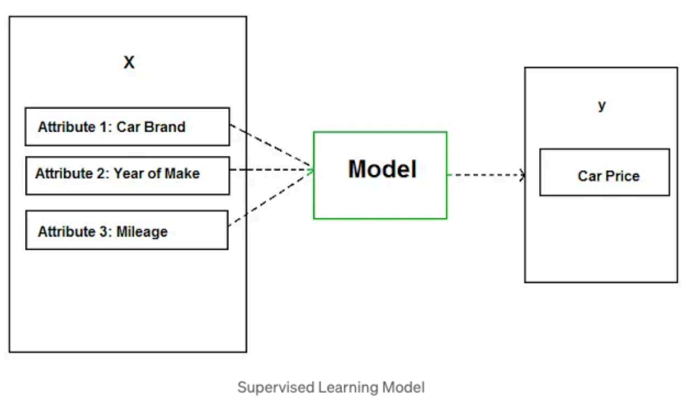

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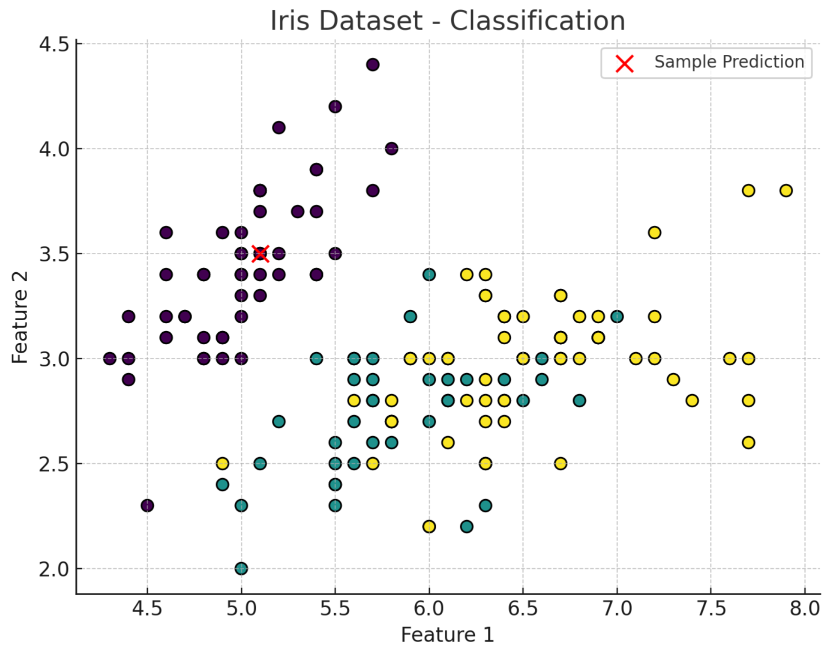


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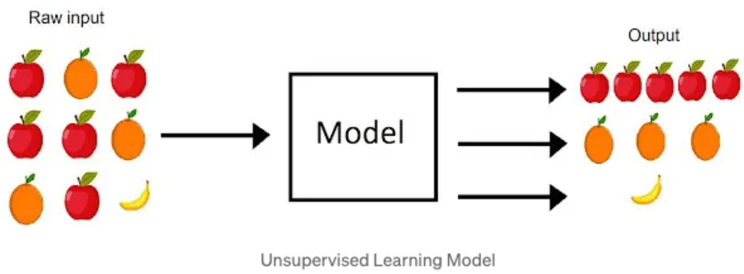

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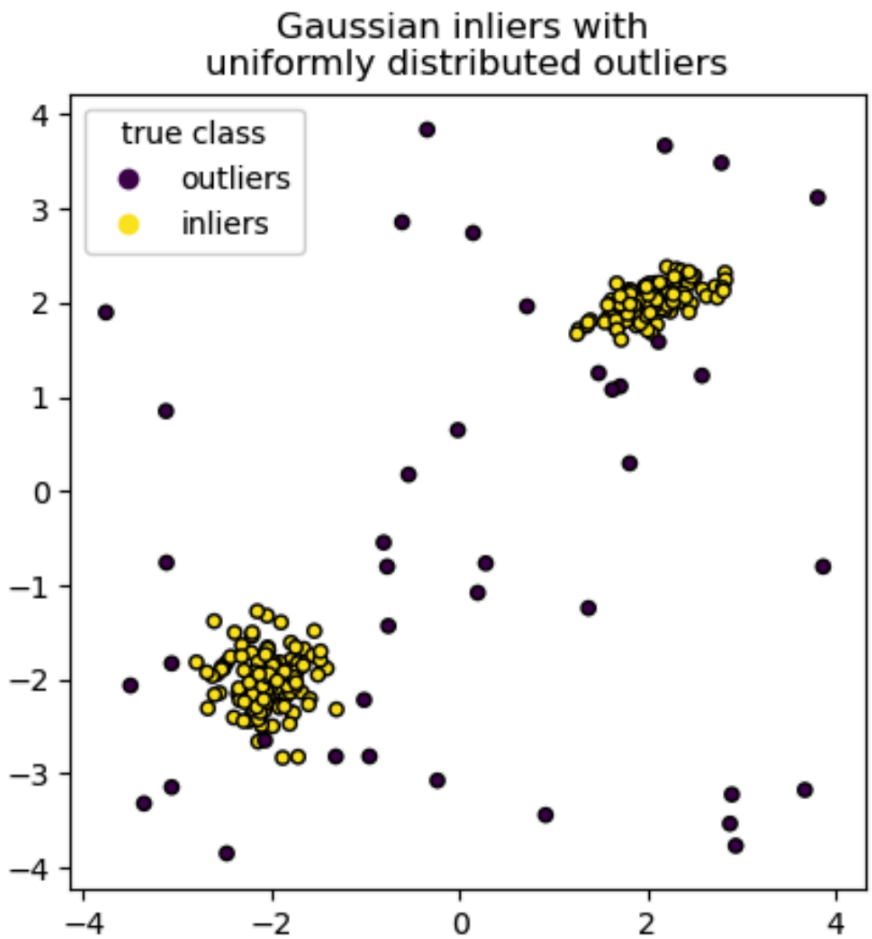

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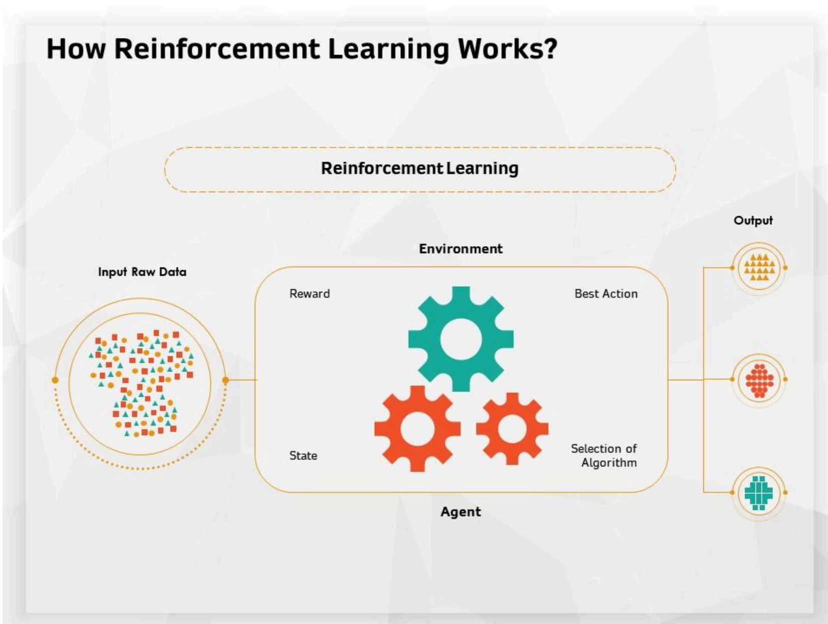

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- **Automated Trading Agent:** Create an RL-based trading bot for stocks or cryptocurrency. The agent makes buy/sell decisions and receives rewards based on the profit or loss of those decisions. Over time, it aims to maximize returns by discovering trading strategies that work well in the market.

- **Traffic Signal Optimization:** Implement an RL agent to control traffic lights in a simulated city grid. The agent gets positive rewards when traffic flow improves (shorter waits, reduced congestion) and negative rewards for long queues or accidents. By learning from these signals, it optimizes the timing of lights to reduce overall traffic jam times.

Real-World Applications

- **Finance**
    - Fraud detection (**Classification, Anomaly Detection**)
    - Stock price prediction (**Regression, Time-Series Forecasting, Reinforcement Learning**)
- **Healthcare**
    - Disease diagnosis (**Classification**)
    - Medical imaging analysis (**Classification, Clustering, Dimensionality Reduction**)
- **Marketing**
    - Recommendation systems (**Clustering, Classification, Association Rule Learning**)
    - Customer segmentation (**Clustering**)
- **Manufacturing**
  Predictive maintenance (**Regression, Anomaly Detection**)
    - Quality control (**Classification, Anomaly Detection**)

Conclusion

Machine learning helps computers learn from data and make predictions without explicit programming. Understanding supervised, unsupervised, and reinforcement learning provides a foundation for applying ML across various industries.